

Table 1: Ablation of SemanticLink components across multiple random seeds

| Model Variant            | Seed           | Accuracy            | Precision           | Recall              | F1-score            | ROC-AUC             | Avg Precision       |
|--------------------------|----------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| Raw text                 | 42             | 0.9227              | 0.8603              | 0.8243              | 0.8419              | 0.9592              | 0.8843              |
|                          | 123            | 0.9300              | 0.8468              | 0.8787              | 0.8624              | 0.9662              | 0.8796              |
|                          | 7              | 0.9415              | 0.8588              | 0.9163              | 0.8866              | 0.9666              | 0.8750              |
|                          | 21             | 0.9342              | 0.8492              | 0.8954              | 0.8717              | 0.9629              | 0.8812              |
|                          | 99             | 0.9310              | 0.8619              | 0.8619              | 0.8619              | 0.9622              | 0.8697              |
|                          | 555            | 0.9289              | 0.8462              | 0.8745              | 0.8601              | 0.9618              | 0.8841              |
|                          | Mean $\pm$ Std | 0.9314 $\pm$ 0.0057 | 0.8539 $\pm$ 0.0066 | 0.8752 $\pm$ 0.0285 | 0.8641 $\pm$ 0.0134 | 0.9632 $\pm$ 0.0026 | 0.8790 $\pm$ 0.0052 |
| SRL Markup               | 42             | 0.9342              | 0.8520              | 0.8912              | 0.8712              | 0.9687              | 0.8892              |
|                          | 123            | 0.9331              | 0.8302              | 0.9205              | 0.8730              | 0.9632              | 0.8686              |
|                          | 7              | 0.9363              | 0.8371              | 0.9247              | 0.8787              | 0.9649              | 0.8815              |
|                          | 21             | 0.9363              | 0.8504              | 0.9038              | 0.8763              | 0.9700              | 0.8972              |
|                          | 99             | 0.9321              | 0.8372              | 0.9038              | 0.8692              | 0.9651              | 0.9005              |
|                          | 555            | 0.9342              | 0.8411              | 0.9079              | 0.8732              | 0.9610              | 0.8747              |
|                          | Mean $\pm$ Std | 0.9343 $\pm$ 0.0015 | 0.8413 $\pm$ 0.0077 | 0.9086 $\pm$ 0.0112 | 0.8736 $\pm$ 0.0031 | 0.9655 $\pm$ 0.0031 | 0.8853 $\pm$ 0.0115 |
| Without Contrastive Loss | 42             | 0.9342              | 0.8577              | 0.8828              | 0.8701              | 0.9622              | 0.8600              |
|                          | 123            | 0.9310              | 0.8264              | 0.9163              | 0.8690              | 0.9599              | 0.8634              |
|                          | 7              | 0.9279              | 0.8400              | 0.8787              | 0.8589              | 0.9630              | 0.8750              |
|                          | 21             | 0.9331              | 0.8302              | 0.9205              | 0.8730              | 0.9585              | 0.8438              |
|                          | 99             | 0.9289              | 0.8577              | 0.8577              | 0.8577              | 0.9689              | 0.8934              |
|                          | 555            | 0.9279              | 0.8455              | 0.8703              | 0.8577              | 0.9643              | 0.8577              |
|                          | Mean $\pm$ Std | 0.9305 $\pm$ 0.0025 | 0.8429 $\pm$ 0.0122 | 0.8877 $\pm$ 0.0231 | 0.8644 $\pm$ 0.0064 | 0.9628 $\pm$ 0.0034 | 0.8655 $\pm$ 0.0155 |
| Without Cross Attention  | 42             | 0.9300              | 0.8554              | 0.8661              | 0.8607              | 0.9698              | 0.8994              |
|                          | 123            | 0.9331              | 0.8405              | 0.9038              | 0.8710              | 0.9671              | 0.8908              |
|                          | 7              | 0.9342              | 0.8667              | 0.8703              | 0.8685              | 0.9674              | 0.8988              |
|                          | 21             | 0.9383              | 0.8571              | 0.9038              | 0.8798              | 0.9660              | 0.8988              |
|                          | 99             | 0.9289              | 0.8406              | 0.8828              | 0.8612              | 0.9666              | 0.8830              |
|                          | 555            | 0.9289              | 0.8519              | 0.8661              | 0.8589              | 0.9646              | 0.8723              |
|                          | Mean $\pm$ Std | 0.9323 $\pm$ 0.0034 | 0.8520 $\pm$ 0.0093 | 0.8821 $\pm$ 0.0163 | 0.8667 $\pm$ 0.0073 | 0.9669 $\pm$ 0.0016 | 0.8905 $\pm$ 0.0100 |